

Cubital Fossa

Short notes - exam-ready, diagram-based anatomy

Definition and importance

The cubital fossa is a triangular hollow on the anterior aspect of the elbow. It is homologous to the popliteal fossa of the lower limb and acts as a passage region for major nerves and vessels travelling from the arm to the forearm.

Clinically it is important for venepuncture, palpation of the brachial pulse, blood pressure recording and injuries around the elbow.

BOUNDARIES AND SHAPE

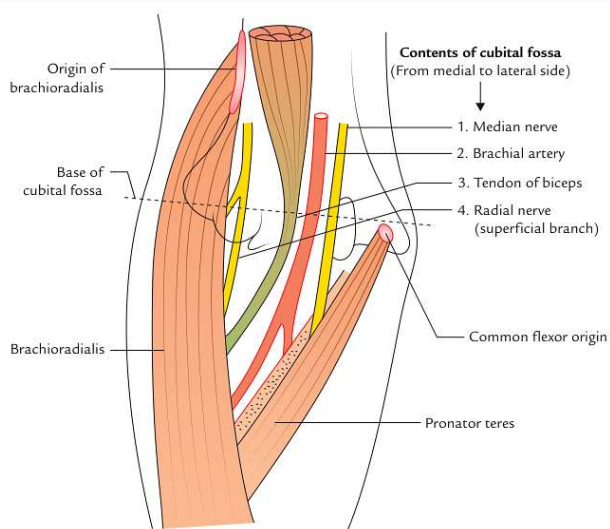


Fig. 8.13 - Boundaries and contents of the cubital fossa.

Triangular limits

Lateral: medial border of brachioradialis

Medial: lateral border of pronator teres

Base: line joining the medial and lateral epicondyles of humerus

Apex: meeting point of brachioradialis and pronator teres

Floor: brachialis above and supinator below

Roof: skin, superficial fascia and deep fascia with bicipital aponeurosis

Diagram rule: draw the muscular borders first, then place the contents from medial to lateral.

Image explanation

The lateral muscular wall is made by brachioradialis and the medial wall by pronator teres. The base is superior, joining the epicondyles, while the apex points inferiorly where the two muscles approach each other. The main contents pass vertically or obliquely through the hollow before entering the forearm.

Shape
triangular depression

Best mnemonic
MBBS

Most used vein
median cubital vein

ROOF AND FLOOR

Floor: deep muscular bed

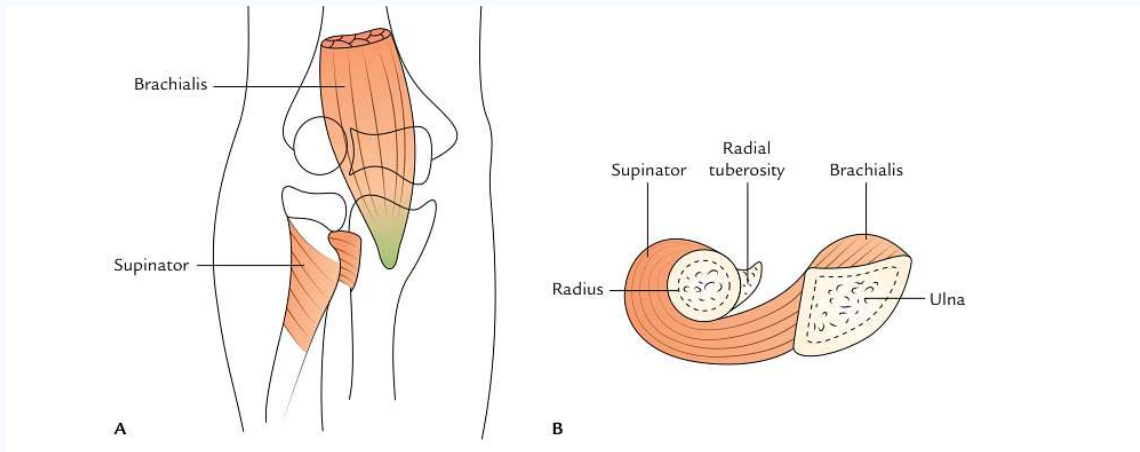


Fig. 8.14 - Muscles forming the floor: brachialis above and supinator below.

Brachialis forms the upper part of the floor and lies deep to the neurovascular contents. Supinator forms the lower part near the radius; this is clinically relevant because the deep branch of radial nerve enters the supinator close to this region.

Roof: superficial venous and fascial layer

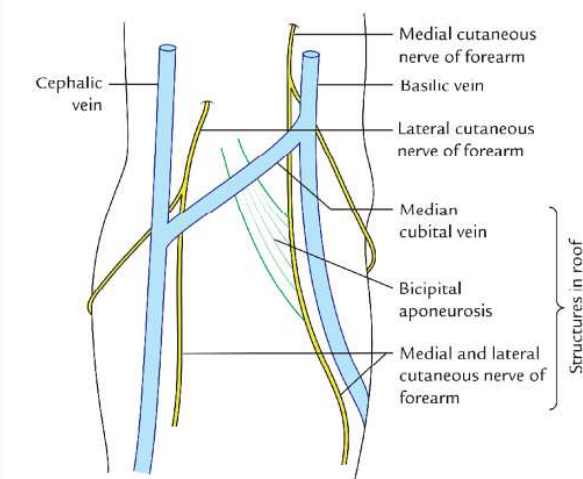


Fig. 8.15 Structures in the roof of cubital fossa.

Fig. 8.15 - Structures in the roof of cubital fossa.

Layers from superficial to deep

1. Skin.
2. Superficial fascia containing median cubital vein, cephalic vein, basilic vein and cutaneous nerves of the forearm.
3. Deep fascia strengthened by bicipital aponeurosis.

The bicipital aponeurosis is an important protective sheet: it separates superficial veins from the brachial artery and median nerve.

Why the roof matters clinically

During venepuncture, the needle enters through the superficial fascia to reach the median cubital vein. The bicipital aponeurosis below the vein reduces the risk of direct injury to the brachial artery and median nerve. This is one reason why the median cubital vein is a preferred site.

CONTENTS AND APPLIED ANATOMY

Contents from medial to lateral - MBBS

M = Median nerve, B = Brachial artery, B = Biceps tendon, S = Superficial radial nerve.

This order is the quickest way to remember the arrangement of the deep structures inside the fossa.

Deep contents

- Median nerve: medial to brachial artery; leaves between two heads of pronator teres.
- Brachial artery: divides into radial and ulnar arteries at the level of neck of radius.
- Biceps tendon: passes backwards and laterally to radial tuberosity.
- Radial nerve: lies between brachialis and brachioradialis; divides into superficial and deep branches.

Clinical points

- Median cubital vein is the commonest site for blood sampling and IV injection.
- Brachial artery is palpated medial to biceps tendon for BP recording.
- Supracondylar fracture may injure the brachial artery and median nerve.
- Deep branch of radial nerve may be compressed near the supinator region.

Venepuncture

Median cubital vein is superficial, stable and supported by bicipital aponeurosis.

BP recording

Brachial pulse is felt medial to biceps tendon; this guides stethoscope placement.

Fracture risk

Brachial artery and median nerve are vulnerable in supracondylar fracture of humerus.

LAST-MINUTE REVISION

5-line exam answer

- Triangular hollow in front of elbow.
- Bounded by brachioradialis, pronator teres and epicondylar line.
- Floor: brachialis and supinator; roof: skin, fascia and bicipital aponeurosis.
- Contents from medial to lateral: MBBS.
- Applied: venepuncture, BP recording, supracondylar fracture.

Viva questions

- Which muscle forms lateral boundary? Brachioradialis.
- Which muscle forms medial boundary? Pronator teres.
- Which artery divides here? Brachial artery.
- Which vein is used for venepuncture? Median cubital vein.
- What protects deep structures? Bicipital aponeurosis.

Source used: Vishram Singh, Textbook of Anatomy: Upper Limb and Thorax, Vol. I, Chapter 8.